National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

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Submitted by:

Afia Maham

Reg number:

23-NTU-CS-1126

Lab no:

14

Semester:

5th

Project Structure:

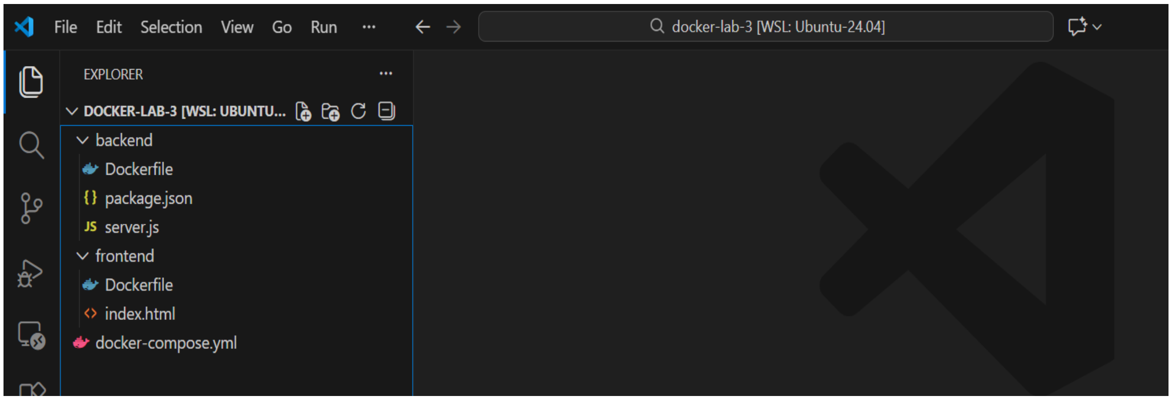


Image creation Frontend:

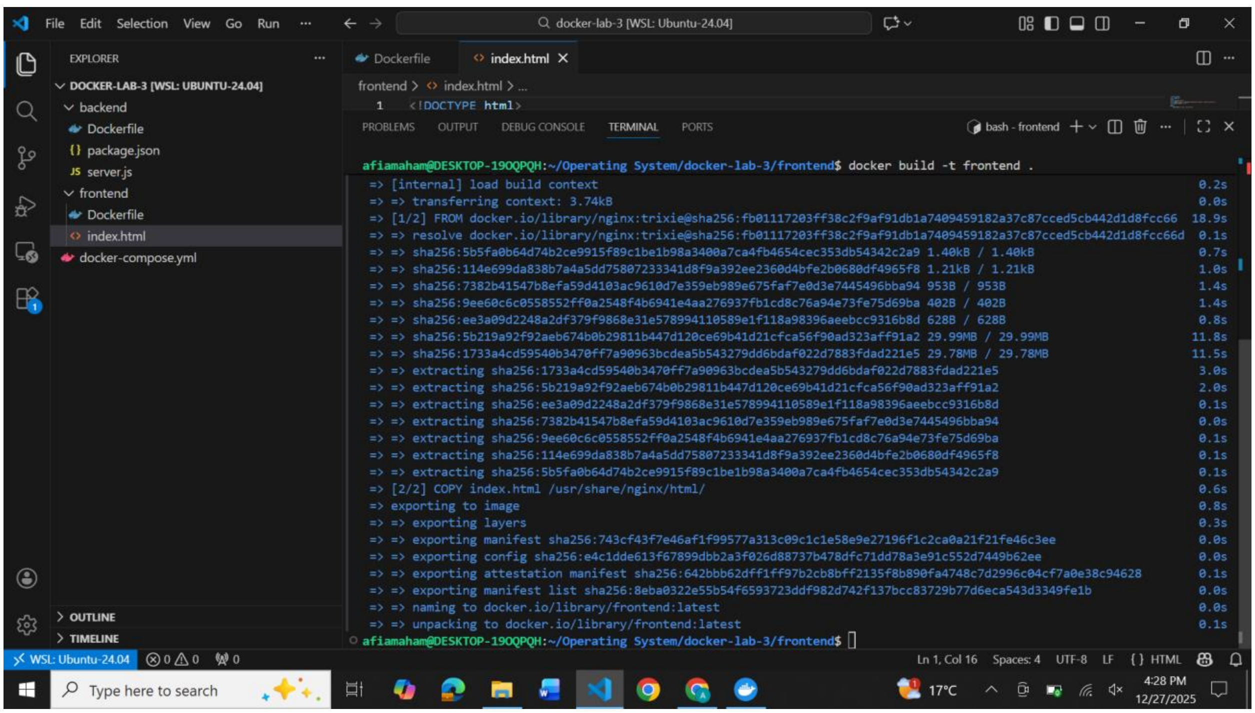
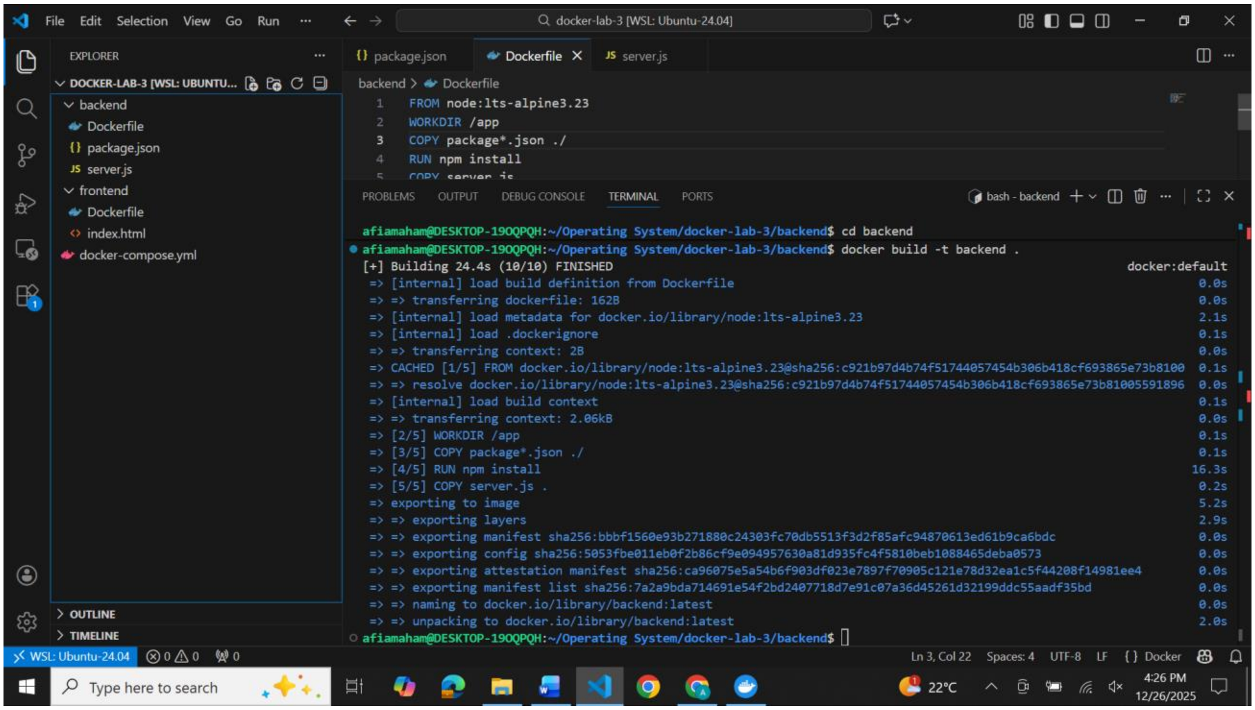
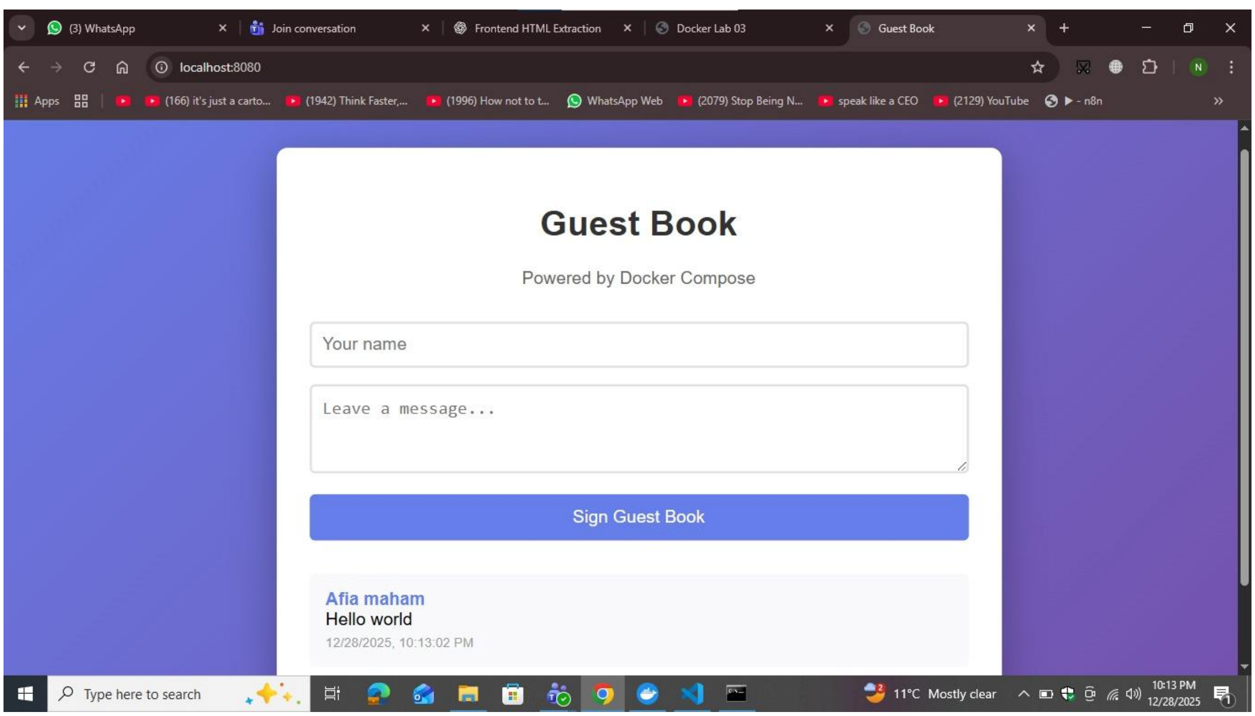


Image creation Backend:



Webpage:



Docker-compose content:

services:

```
  # MongoDB Database
  mongodb:
    image: mongo:7-jammy
    container_name: guestbook-db
    networks:
      - <enter_network>
    volumes:
      - mongo_data:/data/db
# Backend API
api:
  # build: ./backend
  # image: <image_name>
  container_name: guestbook-api
  environment:
    - MONGO_URL=mongodb://mongodb:27017
  ports:
    - "3000:3000"
  depends_on:
    - mongodb
  networks:
    - <enter_network>
# Frontend
web:
  # build: ./frontend
  # image: <image_name>
  container_name: guestbook-web
  ports:
```

```yaml
    - "8080:80"
  depends_on:
    - api
  networks:
    - <enter_network>
networks:
  <enter_network>:
volumes:
  mongo_data:
```

Dockerfile Frontend:

```dockerfile
FROM nginx:trixie
COPY index.html /usr/share/nginx/html/
EXPOSE 80
```

Dockerfile Backend:

```dockerfile
FROM node:lts-alpine3.23
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY server.js .
EXPOSE 3000
CMD ["npm", "start"]
```